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TREND-BASED MOCK PAPER

GATE Electronics and Communication Engineering

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electronics and Communication Engineering
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Sum of first 100 natural numbers:

(Numerical Answer Type — Enter the numerical value)

Q2. [1-MCQ] [1 mark]

Choose opposite of 'benevolent':

- (A) Hostile
- (B) Kind
- (C) Generous
- (D) Helpful

Q3. [1-MCQ] [1 mark]

Choose synonym of 'trivial':

- (A) Significant
- (B) Insignificant
- (C) Important
- (D) Major

Section B: Electronics and Communication Engineering

Q4. [2-MCQ] [2 marks]

NBFM bandwidth approx:

- (A) $2 \cdot f_m$
- (B) $2 \cdot (\Delta f + f_m)$
- (C) Δf
- (D) f_m only

Q5. [2-NAT] [2 marks]

QPSK transmits bits per symbol:

(Numerical Answer Type — Enter the numerical value)

Q6. [2-NAT] [2 marks]

PCM L=16 levels. Bits per sample:

(Numerical Answer Type — Enter the numerical value)

Q7. [2-MCQ] [2 marks]

Nyquist rate for $f_{\max}=4\text{kHz}$:

- (A) 4kHz
 - (B) 8kHz
 - (C) 2kHz
 - (D) 16kHz
-

Q8. [2-MCQ] [2 marks]

NBFM bandwidth approx:

- (A) $2f_m$
 - (B) $2(\Delta f + f_m)$
 - (C) Δf
 - (D) f_m only
-

Q9. [2-NAT] [2 marks]

QPSK transmits bits per symbol:

(Numerical Answer Type — Enter the numerical value)

Q10. [2-MCQ] [2 marks]

Shannon capacity: BW=1MHz, SNR=15. Capacity:

- (A) 4 Mbps
 - (B) 1 Mbps
 - (C) 15 Mbps
 - (D) 5 Mbps
-

Q11. [1-MCQ] [1 mark]

Integral of $1/x$ dx:

- (A) x
 - (B) $\log(x)+C$
 - (C) $1/x^2$
 - (D) $x^{2/2}$
-

Q12. [1-NAT] [1 mark]

$i^4 =$:

(Numerical Answer Type — Enter the numerical value)

Q13. [1-NAT] [1 mark]

$P(A \cap B) = ?$

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q15. [2-NAT] [2 marks]

The Laplace transform of $t^*e^{(-2t)}$ is:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-NAT] [2 marks]

The Laplace transform of $t^*e^{(-2t)}$ is:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

Intrinsic impedance of free space:

- (A) 120π ohm
- (B) 377 ohm
- (C) 50 ohm
- (D) 75 ohm

Q19. [2-MCQ] [2 marks]

$Z_0=50\text{ohm}$, $Z_L=100\text{ohm}$. Reflection coefficient:

- (A) 0.33
- (B) 0.5
- (C) 0.67
- (D) 1.0

Q20. [2-MCQ] [2 marks]

Poynting vector $S =$

- (A) $E \times D$
- (B) $E \times H$
- (C) $H \times E$
- (D) $B \times H$

Q21. [2-MCQ] [2 marks]

Plane wave in $\epsilon_r=4$, $\mu_r=1$. Wavelength:

- (A) $\lambda_0/2$
- (B) λ_0
- (C) $\lambda_0/4$
- (D) $2\lambda_0$

Q22. [2-MCQ] [2 marks]

Skin depth in copper at 1MHz:

- (A) 66 microns
- (B) 66 mm
- (C) 6.6 microns
- (D) 6.6 mm

Q23. [2-MCQ] [2 marks]

4:1 MUX select lines:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q24. [2-MCQ] [2 marks]

$F(A,B,C,D) = \sum m(0,1,2,3,4,5,8,9,10)$. Minimal:

- (A) $A' + B'C$
- (B) $A'D' + A'C'$
- (C) $B' + A'D'$
- (D) $A'B' + A'D'$

Q25. [2-MSQ] [2 marks]

One's complement of 11010110:

- (A) 00101001
- (B) 11010110
- (C) 00101000
- (D) 11111111

Q26. [2-NAT] [2 marks]

4:1 MUX select lines:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MSQ] [2 marks]

Mod-12 counter minimum FFs:

- (A) 3
- (B) 4
- (C) 12
- (D) 8

Q28. [2-MCQ] [2 marks]

FS of $f(t)=t$ on $(-\pi, \pi)$ has only:

- (A) Sine
- (B) Cosine
- (C) Both
- (D) DC only

Q29. [2-MCQ] [2 marks]

Nyquist rate for $f_{\max}=4\text{kHz}$:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 16kHz

Q30. [2-MCQ] [2 marks]

FS of $f(t)=t$ on $(-\pi, \pi)$ has only:

- (A) Sine
- (B) Cosine
- (C) Both

(D) DC only

Q31. [2-MCQ] [2 marks]

Auto-correlation FT =

- (A) $|X(j\omega)|^2$
- (B) $X(j\omega)$
- (C) $S_{xx}(\omega)$
- (D) $\text{Re}[X(j\omega)]$

Q32. [2-MCQ] [2 marks]

Nyquist rate for $f_{\text{max}}=4\text{kHz}$:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 16kHz

Q33. [2-MCQ] [2 marks]

CMRR of diff amp depends on:

- (A) Only R_C
- (B) Only β
- (C) Active load matching
- (D) Temperature

Q34. [2-MCQ] [2 marks]

CC (emitter follower) has:

- (A) High voltage gain
- (B) High current gain
- (C) Low input Z
- (D) Voltage gain > 1

Q35. [2-MCQ] [2 marks]

Input offset voltage causes:

- (A) Output saturates
- (B) Output DC offset
- (C) No effect
- (D) BW increases

Q36. [2-NAT] [2 marks]

Input offset voltage causes:

(Numerical Answer Type — Enter the numerical value)

Q37. [1-NAT] [1 mark]

Maxwell has how many equations:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-NAT] [2 marks]

Characteristic impedance $Z_0 =$:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q40. [2-MCQ] [2 marks]

Skin depth increases with:

- (A) Frequency
- (B) Conductivity
- (C) Both
- (D) Neither

Q41. [1-MCQ] [1 mark]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q42. [1-MCQ] [1 mark]

;

- (A) ; $\partial f^{3\pm}$
- (B) ; $\partial f^{0\pm}$
- (C) I_c/I_b
- (D) I_e/I_c

Q43. [1-MCQ] [1 mark]

Ideal op-amp open-loop gain:

- (A) 0
- (B) 1
- (C) Infinite
- (D) -1

Q44. [1-MCQ] [1 mark]

FET is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q45. [1-MCQ] [1 mark]

RC oscillator frequency:

- (A) $f=1/(2\pi RC)$
- (B) $f=1/RC$
- (C) $f=1/(4\pi RC)$
- (D) $f=2\pi RC$

Q46. [1-MCQ] [1 mark]

n-type majority carriers:

- (A) Holes
- (B) Electrons
- (C) Both equal
- (D) None

Q47. [1-MCQ] [1 mark]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q48. [1-NAT] [1 mark]

SCR has:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Zener diode used for:

- (A) Rectification
- (B) Voltage regulation
- (C) Amplification
- (D) None

Q50. [1-MCQ] [1 mark]

4-bit binary counter counts:

- (A) 0-15
- (B) 0-9
- (C) 1-16
- (D) 1-10

Q51. [1-MCQ] [1 mark]

D flip-flop follows input after:

- (A) Same instant
- (B) One clock
- (C) Two clocks
- (D) Never

Q52. [1-MCQ] [1 mark]

ROM is:

- (A) Read only
- (B) Read write
- (C) Volatile
- (D) Cache

Q53. [1-NAT] [1 mark]

TTL stands for:

(Numerical Answer Type — Enter the numerical value)

Q54. [1-MCQ] [1 mark]

Routh-Hurwitz determines:

- (A) Stability
- (B) Time response
- (C) Frequency
- (D) Sensitivity

Q55. [1-MCQ] [1 mark]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q56. [1-MCQ] [1 mark]

Nyquist criterion uses:

- (A) Polar plot
- (B) Root locus
- (C) Routh
- (D) Bode

Q57. [1-MCQ] [1 mark]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q58. [1-MCQ] [1 mark]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q59. [1-MCQ] [1 mark]

PCM stands for:

- (A) Pulse Code Modulation
- (B) Phase Code Modulation
- (C) Pulse Carrier Modulation
- (D) None

Q60. [1-MCQ] [1 mark]

FM has:

- (A) Constant amplitude
- (B) Variable
- (C) Both
- (D) None

Q61. [1-MCQ] [1 mark]

ASK modulation varies:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q62. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) $e^{-3/s}$

Q63. [2-MCQ] [2 marks]

Wheatstone bridge null condition:

- (A) $R_1/R_2 = R_3/R_4$
- (B) $R_1 \cdot R_4 = R_2 \cdot R_3$
- (C) Both A and B
- (D) $R_1 + R_2 = R_3 + R_4$

Q64. [1-MCQ] [1 mark]

Z-parameter conditions:

- (A) Open circuit
- (B) Short circuit
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

h-parameter represents:

- (A) Open circuit
 - (B) Short circuit
 - (C) Hybrid
 - (D) Inverse
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1MCQ	B	1
5	1NAT	B	1
6	1NAT	B	1
7	1MCQ	A	1
8	1MCQ	C	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1NAT	B	1
12	1MCQ	B	1
13	1MCQ	B	1

14	1MCQ	C	1
15	1MCQ	C	1
16	1MCQ	B	1
17	1MCQ	A	1
18	1MCQ	A	1
19	1MCQ	B	1
20	1MCQ	A	1
21	1NAT	A	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1MCQ	A	1
25	1MCQ	B	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1MCQ	A	1

29	1MCQ	A	1
30	1NAT	C	1
31	2NAT	A	2
32	2MCQ	B	2
33	2MCQ	D	2
34	2MCQ	B	2
35	2MCQ	A	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2MCQ	C	2
41	2MCQ	B	2
42	2MCQ	B	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	B	2
46	2MSQ	A	2
47	2NAT	B	2
48	2MCQ	B	2
49	2MCQ	C	2
50	2MCQ	B	2
51	2NAT	B	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2NAT	B	2
55	2NAT	A	2
56	2MSQ	B	2
57	2MCQ	A	2
58	2NAT	B	2

59	2NAT	B	2
60	2NAT	A	2
61	2MCQ	A	2
62	2MCQ	A	2
63	2MCQ	B	2
64	2NAT	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: difficult

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Complex Analysis | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Network, Signals & Systems | Topic: Two Port | Difficulty: easy

Q8.

Answer: C

Solution:

Correct Answer: C

Subject: Network, Signals & Systems | Topic: Two-Port Networks | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Semiconductors | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: MOSFET | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: SCR | Difficulty: easy

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Diodes | Difficulty: easy

Q13.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: Amplifiers | Difficulty: easy

Q14.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: BJT | Difficulty: difficult

Q15.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: Op-Amps | Difficulty: moderate

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: FET | Difficulty: moderate

Q17.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Circuits | Topic: Oscillators | Difficulty: easy

Q18.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Counters | Difficulty: moderate

Q19.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Circuits | Topic: Flip-Flops | Difficulty: moderate

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Memory | Difficulty: moderate

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Logic Families | Difficulty: moderate

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: moderate

Q23.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: difficult

Q24.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Transfer Function | Difficulty: easy

Q26.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Modulation | Difficulty: moderate

Q27.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital Comm | Difficulty: difficult

Q28.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Analog | Difficulty: difficult

Q29.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital | Difficulty: moderate

Q30.

Answer: C

Solution:

Correct Answer: C

Subject: Electromagnetics | Topic: Maxwell Equations | Difficulty: easy

Q31.

Answer: A

Solution:

Correct Answer: A

Subject: Electromagnetics | Topic: Transmission Line | Difficulty: difficult

Q32.

Answer: B

Solution:

Correct Answer: B

Subject: Electromagnetics | Topic: Waveguides | Difficulty: difficult

Q33.

Answer: D

Solution:

Correct Answer: D

Subject: Electromagnetics | Topic: Wave Propagation | Difficulty: easy

Q34.

Answer: B

Solution:

$\eta_0 = \sqrt{\mu_0/\epsilon_0} = 377 \text{ ohm approx } 120\pi$.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q35.

Answer: A

Solution:

NBFM: BW approx $2f_m$ (same as AM). WBFM: Carson rule.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q36.

Answer: B

Solution:

4:1: $\log_2(4) = 2$ select lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q37.

Answer: A

Solution:

$\Gamma = (100-50)/(100+50) = 50/150 = 1/3 = 0.33$.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q38.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

Odd function: only sine terms.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q40.

Answer: C

Solution:

CMRR depends on active load matching. Active load improves CMRR.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q41.

Answer: B

Solution:

$S = E \times H$. Direction: power flow. Magnitude: power per unit area.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q42.

Answer: B

Solution:

$f_s \geq 2f_{\max} = 8\text{kHz}$.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q43.

Answer: A

Solution:

$v = c/\sqrt{4 \times 1} = c/2$. $\lambda = v/f = \lambda_0/2$.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q44.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q45.

Answer: B

Solution:

K-map: $F = A'(D' + B'C') = A'D' + A'B'C'$. Closest: $A'D' + A'C'$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q46.

Answer: A

Solution:

Flip all bits: 11010110 -> 00101001.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q47.

Answer: B

Solution:

QPSK: 4 phase states. $\log_2(4) = 2$ bits per symbol.

Subject: Communication | Topic: Communication | Difficulty: easy

Q48.

Answer: B

Solution:

CC: $A_v \sim 1$, high input Z, high current gain, low output Z.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q49.

Answer: C

Solution:

Null: $R_1/R_2 = R_3/R_4$, which implies $R_1 \cdot R_4 = R_2 \cdot R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q50.

Answer: B

Solution:

Input offset: differential DC input causing DC output offset.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q51.

Answer: B

Solution:

$L = 2^n$. $n = \log_2(16) = 4$ bits.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q52.

Answer: A

Solution:

Odd function: only sine terms.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: easy

Q53.

Answer: B

Solution:

$f_s \geq 2 \cdot 4\text{kHz} = 8\text{kHz}$.

Subject: Communication | Topic: Communication | Difficulty: easy

Q54.

Answer: B

Solution:

4:1: $\log_2(4) = 2$ select lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q55.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

$2^n \geq 12$. $n=4$ gives 16 states. Minimum 4 FFs.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: difficult

Q57.

Answer: A

Solution:

NBFM: BW approx $2f_m$ (same as AM). WBFM: Carson rule.

Subject: Communication | Topic: Communication | Difficulty: difficult

Q58.

Answer: B

Solution:

QPSK: 4 phase states. $\log_2(4) = 2$ bits per symbol.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q59.

Answer: B

Solution:

Input offset: differential DC input causing DC output offset.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q60.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: A

Solution:

Wiener-Khinchin: FT of $R_{xx}(\tau) = |X(j\omega)|^2$.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: easy

Q62.

Answer: A

Solution:

$\delta = \sqrt{2/(\omega\mu\sigma)}$. At 1MHz, δ approx 66 microns.

Subject: EMFT | Topic: EMFT | Difficulty: easy

Q63.

Answer: B

Solution:

$f_s \geq 2f_{\max} = 8\text{kHz}$.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: difficult

Q64.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q65.

Answer: A

Solution:

$C = B \log_2(1+\text{SNR}) = 1\text{e}6 \log_2(16) = 1\text{e}6 \cdot 4 = 4\text{ Mbps}$.

Subject: Communication | Topic: Communication | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Communication: 7 questions, 14 marks
Engineering Mathematics: 7 questions, 11 marks
EMFT: 5 questions, 10 marks
Digital Electronics: 5 questions, 10 marks
Signal Systems: 5 questions, 10 marks
Analog Electronics: 4 questions, 8 marks
Electromagnetics: 4 questions, 7 marks
Analog Circuits: 5 questions, 5 marks
Electronic Devices: 4 questions, 4 marks
Digital Circuits: 4 questions, 4 marks
Control Systems: 4 questions, 4 marks
Communication Systems: 4 questions, 4 marks
Network Theory: 2 questions, 4 marks
General Aptitude: 3 questions, 3 marks
Network, Signals & Systems: 2 questions, 2 marks

Difficulty Distribution:

Easy: 24 | Moderate: 30 | Difficult: 11

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.